Christian Dane Beels

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Education

United States Military Academy — West Point, NY
B.S. in Mathematical Sciences — Expected 2028
GPA: 4.007

Research interests: Cybersecurity; artificial intelligence; computational mathematics.

Research Experience

Topological Data Analysis for Data Poisoning Prevention

Cadet Researcher — *Jan. 2026 – Present*

Department of Mathematical Sciences, USMA | Advisor: Dr. Joseph Dorta

- Reconstruct and modify a topological data analysis algorithm to investigate detection and removal of poisoned records from network intrusion detection system training data.
- Paper accepted for presentation at the 2026 MIT Undergraduate Research Technology Conference (URTC); unable to attend.
- Received the Outstanding Poster Award at MAA MathFest 2026.

Hunting Hackers with Data

Cadet Researcher — *Aug. 2026 – Present*

Department of Mathematical Sciences, USMA | Advisor: CPT Nolan Hedglin

- Explore data analysis methods to identify adversarial activity on computer networks using data supplied by the Cyber Protection Brigade.

Speech Recognition as a Threat Vector in Real-Time Doxxing

Cadet Researcher — *Sep. 2025 – Dec. 2025*

Army Cyber Institute, West Point, NY | Advisor: MAJ Nicholas Harrell, PhD

- Studied the feasibility of adversaries using consumer wearable technology to expose information on U.S. forces in real time.
- Authored a paper on methods, findings, and operational security implications; presented findings to the USMA Department of Electrical Engineering and Computer Science.

Radiation Environment Modeling for Interceptor Survivability

Research Project Intern — *June 2025 – July 2025*

Missile Defense Agency, Redstone Arsenal, AL | Advisor: Michael McKenna

- Developed and analyzed nuclear radiation environment models to inform survivability and lethality considerations for anti-ICBM interceptor launch scenarios.
- Presented models and results to a panel of agency and military leadership.

SPICE-Based Modeling of Space Radiation Effects on Satellite Amplifiers

Cadet Researcher — *Jan. 2025 – Apr. 2025*

Department of Physics and Nuclear Engineering, USMA | Advisor: LTC Jill Rahon, PhD

- Modeled space radiation effects on RF amplifiers using SPICE simulations to evaluate performance degradation and identify conditions associated with loss of signal integrity.

Presentations

Beels, C. D. (Aug. 2026). *Addressing a Blind Spot in a Topological Data Analysis–Based Data Poisoning Detector.* Poster presentation, MAA MathFest. Outstanding Poster Award.

Beels, C. D. (May 2026). *Topological Data Analysis for Data Poisoning Prevention in Network Intrusion Detection Systems.* Poster presentation, USMA Math Research Symposium (MaRS).

Beels, C. D. (Dec. 2025). *Speech Recognition as a Threat Vector in Real-Time Doxxing.* Poster presentation, USMA Department of Electrical Engineering and Computer Science.

Beels, C. D., & Ruffing, C. (Apr. 2025). *SPICE-Based Modeling of Space Radiation Effects on Satellite Amplifiers.* Poster presentation, USMA Department of Physics and Nuclear Engineering.

Honors and Awards

Outstanding Poster Award, MAA MathFest	2026
Distinguished Cadet Award, USMA (*Top 5% Academics in Class*)	2026
Master of the Sword Award, USMA (*>A- Physical Program Score*)	2026
Superintendent's Award, USMA (*>3.5 average across Academic, Military, Physical Pillars*)	2025
Dean's List, USMA (four recognitions)	2025, 2026
Norwegian Foot March Badge (Silver)	2025, 2026

Technical Skills

Programming	Python, C, HTML, x86-64 Assembly
Scientific computing	giotto-tda, PyTorch, NumPy, Matplotlib, Wolfram Mathematica
Modeling	SPICE (Simulation Program with Integrated Circuit Emphasis)

Leadership and Military Experience

Assistant Cadet-in-Charge, West Point Cyber Team Aug. 2025 – Present
Department of Electrical Engineering and Computer Science, USMA

- Lead team participation in collegiate, national, and international cybersecurity competitions; ranked in the top 1% individually and top 0.5% with the team in National Cyber League; qualified for President's Cup Finals, Individual Offensive Track.

Headquarters Platoon Sergeant, USMA Company G-1 Aug. 2026 – Dec. 2026

- Responsible for the success and accountability of more than 20 subordinate staff officers.

Cadet Basic Training Squad Leader, Hotel Company Jun. 2026 – Jul. 2026

- Led and mentored nine new cadets; taught military tactics, rifle marksmanship, land navigation, and character development.

Team Leader, USMA Company G-1 Aug. 2025 – May 2026

- Mentored an underclass cadet in academic, military, physical, and character development.

Air Assault School, 101st Airborne Division June 2025
Trained in helicopter operations, sling-load operations, and rappelling.

Teaching, Service, and Outreach

Tutor, West Point Center for Enhanced Performance 2024 – Present
Lead evening study sessions before major graded events in single- and multivariable calculus, computer science, chemistry, psychology, and economics.

Volunteer Web Developer, ProVetUSA 2022 – 2023
Built online course materials to support veterans transitioning to civilian employment.

Volunteer, Fuller Center for Housing of Greater NYC 2020 – 2024
Contributed to construction of low-income housing in Westchester, NY.